

Report on

“Design & construction of Power Transformer”

Seminar

Speaker

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Submitted to

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In this session we learn about the topic Introduction to Instrument Transformers and Substation Equipment, which is very important for understanding high voltage power systems. At the beginning of the session, Sir discussed the importance of transformers in power systems and mentioned that some of the biggest power transformers in the world are manufactured in Japan and China where very large-capacity transformers are used in high-voltage transmission networks to handle huge power levels efficiently. The main objective of this session was to learn why instrument transformers are used and how different substation equipment helps in safe and reliable power transmission and distribution. At first, the teacher explained that in power systems, very high voltage and current cannot be measured directly using normal meters because it is unsafe and impractical. Current Transformers (CT) step down high current to 1A or 5A and Potential Transformers (PT) reduce high voltage to a safe value like 110V. The importance of IEC standards such as IEC 61869 was explained which ensure safety and reliability of electrical equipment worldwide. The session also covered indoor Substations which are used in urban areas for better protection and safety and Outdoor Substations which are used for high-voltage systems because they are cost effective and easy to expand. Specialized protection and regulation are achieved through Buchholz Relays, Pressure Release Devices, and Tap Changers, including On-Load Tap Changers and Off-Load variants that adjust voltage in precise increments. Substation design distinguishes between gas insulated using Sulfur Hexafluoride and Air Insulated systems noting that while indoor setups increase costs they allow for modern compact designs. Battery Banks are essential in substations to provide DC power for control systems as AC cannot be stored and would fail during a power outage. Technical design aspects such as oil injection from the bottom to eliminate bubbles and the use of vector group symbols like Dyn11 to indicate HV-LV phase displacement are critical for ensuring operational

reliability. The teacher also explained the types of transformation, such as step-up transformation at generating stations to reduce transmission losses and step-down transformation at receiving and distribution substations to supply power to consumers at usable voltage levels. Sir not only explained the technical concepts but also shared practical knowledge for career. Sir talked about the current job market condition, explaining that there is strong demand for engineers with good basic concepts, practical understanding, and knowledge of standards. He guided us on how to prepare for a job interview with clear fundamentals, confidence, communication skills, and practical examples. Sir also shared some common viva questions, such as the difference between CT and PT, why CT secondary should not be open, purpose of substations, and role of IEC standards, which helped us understand how to prepare for viva and interviews. After everything when it comes to the QNA session Many students ask different based on his doubt but one question was that on my favorite that The basic transformer color we can see outside it's It looked like a gray color and white colored inside we all know color was not playing a vital role here So Why anyone Change the color? Here Sir responded that it can be any color the manufacturer wants or whatever demand wants. It will black,pink,blue,green any type of the color doesn't matter here in this case. Then Sir told us about some industries names who manufacture the transformer in different colors like green or purple. Overall this session was very informative as it combined technical learning, real-world industry knowledge, career guidance giving us a clear picture of both academic and professional aspects of power engineering.